Lam Fung Academy · LF Academy

Primary 5 · 20th Lesson · Student Handout

Simulation 2 Review + Winter Vacation SSPA Special Training Plan

Semester 1 Final Lesson · Review and Reflection · 75 minutes · 1-to-3 Online Lesson

Corresponding textbooks: L19 SSPA Mock Exam (2) Review + Winter Vacation Independent Practice Plan

Core Traps: □ T1 Multi-digit · T2 Decimal · T3 Reverse Questions · T5 Area · T9 Score - Simulation 2 Five Major Score-Losing Areas

SSPA Related: 🔴 High frequency Integrated diagnosis of sub-test traps, personalized weakness location

Pre-requisite knowledge: L1–L18 full semester content + L19 mock exam

Objective of this class: ❶ Diagnose the reasons for losing points in the five major traps in Simulation 2 ❷ Self-assessment of personal weaknesses ❸ Develop a 15-minute daily exercise plan during winter vacation ❹ Targeted error correction exercises

Student Name:	Class:	Date:	Time Spent:

I.Mock Exam2 SSPA five major trap disaster areas (L19 review)

Trap 1: T1 multi-digit number - "rounding" approximation trap 🔴 SSPA

① When taking approximate values, you must clearly see "which digit to take" - tens of thousands, thousands, hundreds?
② Key rules: Look at the target position**The next digit**The number determines whether to "enter" or "round"
③ Common mistakes: when you get to the "ten thousand digits", you stop after looking at the thousands digits, without confirming the value of the tens of thousands digits
④ Another mistake: forgetting to clear all subsequent digits after placing them (for example, 345,678 rounded to the nearest ten thousand = 350,000, not 35,000)

Example questions T1-1: Similar questions to Question 5 of Simulation 2

Round 2,847,563 to the nearest hundred thousand digits.

❌ Common mistakes

2,800,000

I only looked at 4 (round) in the tens of thousands place, but the hundreds of thousands place is 8. I need to look at the next level in the millions place.

✅ Correct solution

2,800,000

The hundred thousand digit = 8, look at the ten thousand digit = 4 (<5, round down) → 2,800,000

Trap 2: T2 Decimal — Decimal Point Position Trap for Decimal Multiplication 🔴 SSPA

① Multiply decimals: first multiply the integers, then count the total number of decimal places and add the decimal point
② **The most frequent error |||SEP|||: 0.3 × 0.4 = 1.2 (correct = 0.12) - Forgetting that multiplying decimals by decimals will get smaller and smaller**③ The second error: the decimal point is placed in the wrong position - it should be from the far right of
|||SEP||| Start counting to the left④ **Verification formula: "Multiplier < 1 → Product < Multiplicand" (0.3 × 0.4 → The answer must be < 0.3)**Start counting to the left
④ Verification formula: "Multiplier < 1 → product < multiplicand" (0.3 × 0.4 → the answer must be < 0.3)

Example questions T2-1: Similar questions to Question 2 of Simulation 2

Calculate 0.25 × 0.4 = ?

❌ **Common mistakes**

$0.25 \times 0.4 = 1.00$

Directly 25×4=100, I forgot to count from the right at the decimal point

✅ **Correct solution**

$0.25 \times 0.4 = 0.100 = 0.1$

25×4=100, the total number of decimal places is 2+1=3 → 0.100 → the simplest 0.1

Pitfall 3: T5 Area — Trapezoidal/Polygonal Area Unit Confusion 🔴 SSPA

① Area of trapezoid = (upper base + lower base) × height ÷ 2 ——**not**(upper base + lower base + height) × 2

② Area of parallelogram = base × height (|||SEP||| not**base × hypotenuse)**③ Area of triangle = base × height ÷ 2 —— Must divide |||SEP|||

③ Area of triangle = base × height ÷ 2 —— required**divide by 2** !

④ Combined graphics: first divide into basic graphics, find the area respectively, and then add the sum

Example T5-1: Simulation 2 trapezoid area problem

The upper base of the trapezoid is 8 cm, the lower base is 12 cm, and the height is 5 cm. Area = ?

❌ **Common mistakes**

$(8+12+5) \times 2 = 50\ cm^2$

Confusing the perimeter and area formulas - adding the height and multiplying by 2

✅ **Correct solution**

$(8+12)\times5\div2 = 50\ cm^2$

(Upper bottom + Lower bottom)×Height÷2 = 20×5÷2 = 50 cm^2

Trap 4: T9 fractions - addition and subtraction with different denominators "straight addition and straight subtraction" 🔴 SSPA

① Fractions with different denominators**cannot**direct addition and subtraction - they must be divided first

② Three steps: find LCM → expand the numerator simultaneously → add the numerators with the same denominator

③ The answer must be reduced to**The simplest fraction |||SEP|||, the improper fraction must be transformed**Mixed fractions④ **Adding and subtracting mixed fractions: Recommended to convert to improper fractions (least error-prone)**

④ Adding and subtracting mixed numbers: It is recommended to convert to improper fractions (least error-prone)

Example T9-1: Simulation 2 Fraction Questions

$\frac{2}{3} + \frac{1}{4} = ?$

❌ **Common mistakes**

$\frac{3}{7}$

✅ **Correct solution**

$\frac{11}{12}$

Numerator + numerator, denominator + denominator - different denominators can never be added directly!

LCM(3,4)=12 → $\frac{8}{12}+\frac{3}{12}=\frac{11}{12}$

Trap 5: T3 inverse question - find the original number when the result is known 🔴 SSPA

① The key to the reverse question: start from the final result of|||SEP||| **Auxiliary: Assume the unknown → Column → Solve the equation**④ Verification: Substitute the answer back to the original question to check whether it is reasonable**Push back**Operation
② When working backwards, addition and subtraction are reciprocal, and multiplication and division are reciprocal: the original "addition" becomes "subtraction" when working backwards.
③ Best to use**equation**Auxiliary: Assume unknowns → List → Solve equations
④ Verification: Substitute the answer back to the original question to check whether it is reasonable

Example T3-1: Simulation 2 reverse question

Xiao Ming has some candy. After taking |||SEP|||, I bought 15 more pills and now I have 45 pills. How many grains were there?$\frac{1}{3}$Later, I bought 15 more pills, and now I have 45 pills. How many grains were there?

❌ Common mistakes

$$45 + 15 = 60 \cdot 60 \times \frac{1}{3} = 20$$

Calculate forward, not backward - you bought 15 before reaching 45, you should subtract 15 first

✅ Correct solution

$$(45 - 15) \div (1 - \frac{1}{3}) = 30 \div \frac{2}{3} = 45$$

Working backward: there were 30 pills before buying 15, this is the remaining 23 after eating 13

🧠 General tips for the five major traps: "Looking at multiple digits, decimal points, memorizing area formulas, dividing fractions first, and working backwards"

II. Self-Assessment Checklist of Personal Weaknesses (Self-Assessment Checklist)

⚠️ **Please honestly self-assess the following 10 items and check the ones where you often make mistakes. This is the "bullseye" for your winter vacation training!**

✓	Weakness description	Trap category	SSPA frequency
☐	**Q1.**When approximating multiple digits, the wrong target digit is taken (for example, rounding to the thousands digit is regarded as rounding to the	T1 multi-digit	🔴 high

	10,000 digit)		
☐	**Q2.**After approximating multiple digits, forget to clear the following digits to zero.	**T1 multi-digit**	🔴 high
☐	**Q3.**Wrong decimal point position in decimal multiplication (especially the 0.3 × 0.4 category)	**T2 decimal**	🔴 high
☐	**Q4.**The decimal point position of the quotient of decimal division is wrong	**T2 decimal**	🟡 medium
☐	**Q5.**Forgot to divide the area of the trapezoid by 2, or use the wrong formula (for example, add the height and then multiply)	**T5 area**	🔴 high
☐	**Q6.**Area of combined graphics: Forgot to divide or overlapping parts are calculated incorrectly	**T5 area**	🔴 high
☐	**Q7.**Direct addition and subtraction of fractions with different denominators (numerator + numerator, denominator + denominator)	**T9 score**	🔴 high
☐	**Q8.**I forgot to reduce the fraction answer, or the improper fraction was not converted into a mixed number.	**T9 score**	🔴 high
☐	**Q9.**When working backwards in the reverse problem, the direction of operation is reversed (when adding, subtracting, forgetting to change)	**T3 reverse**	🔴 high
☐	**Q10.**Missing answer sentences for application questions/Missing units/Confusing columns	**T7 Application**	🟡 medium

III.Winter SSPA per day 15 minutes special training plan (two weeks = 14 days)

Planning Principles 🔴 SSPA Preparation

① Only**15 minutes**——The focus is**continuous**rather than long-term

② Focus on**1 trap |||SEP|||, practice the five major traps in turn (can be cycled twice in two weeks)**Each exercise: 3 minutes to review the rules → 10 minutes to do the questions → 2 minutes to mark the answers

④ Rest on Sunday (or make up for unfinished exercises during the week)

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day	date	focus trap	Exercise content (15 minutes)	✓
1	Day 1	T1 multi-digit	5 questions on taking approximate values + 5 questions on reading and w riting numbers	☐
2	Day 2	T2 decimal	5 questions on multiplication of decimals + 5 questions on division of decimals	☐
3	Day 3	T5 area	3 questions on the area of a trapezoid + 3 questions on the area of a triangle + 2 questions on combined figures	☐
4	Day 4	T9 score	5 questions on adding and subtracting different denominators + 5 questions on adding and subtracting mixed fractions	☐
5	Day 5	T3 reverse	5 inverse problems (w ith equation assistance) + 3 application problems	☐
6	Day 6	T1+T2 mixed	Multi-digit + decimal mixed exercises 10 questions (timed 12 minutes)	☐
7	Day 7	🌿 **Rest days (or make up for unfinished questions from Day 1-6)**		☐
8	Day 8	T5 area	10 questions on mixed areas of all shapes (timed 12 minutes)	☐

9	Day 9	T9 score	Four mixed fractions 5 questions + 5 application questions	☐
10	Day 10	T3 reverse	5 questions on converse problems + 5 questions on application of equations	☐
11	Day 11	T2 decimal	10 mixed decimal questions (including approximate values)	☐
12	Day 12	comprehensive	Selected 10 questions from the mock paper (limited to 15 minutes)	☐
13	Day 13	Personally the weakest	For the traps with the most incorrect marks on Days 1-12, do 10 more questions.	☐
14	Day 14	📋 **Summary: Count the wrong questions in the past two weeks and make a list of "still need to improve"**		☐

⚠️ **Key rules: After each exercise, you must "correct the answer + mark the wrong question + understand the cause of the error." Doing nothing = doing nothing.**

IV.L19 Common Mistake for correctness reviewquestion (total 20 question · Focus on the five traps)

Group A — T1 morenumber of digitsapproximate value (questions 1-4)

📟 **Exam tips (SSPA must read)**

① Do the questions you know first, don't get stuck on one!
② Check the unit for each question (cm vs cm² vs cm³)!
③ Geometry questions: If there is an elevation in the picture, use the height. If there is no elevation, use the formula to find it!
④ Application question: Write an answer sentence! Points will be deducted for not writing steps!
⑤ 5 minutes left: Check if the MC has been filled in and if the unit is correct!

#	Question	Difficulty	Working Space
R1	Approximate 3,456,789 to the "hundredthousandth digit".	🌱	
R2	Round 67,890,123 to the nearest million.	🌿	
R3	Round 4,507,632 to the nearest ten thousand. And write which digit you are looking at.	🌿	
R4	Which of the following is a correct approximation to the millionth place of 29,876,543? A. 29,000,000 B. 30,000,000 C. 29,900,000 D. 20,000,000	🌳	

Group B — T2 decimalcalculate (questions 5-8)

#	Question	Difficulty	Working Space
R5	Calculate 0.6 × 0.3 = ?	🌱	
R6	Calculate 0.25 × 0.8 = ?	🌿	
R7	Calculate 0.15 × 0.4 = ? (Write the correct decimal point position after 15×4=60)	🌿	
R8	Calculate 0.05 × 0.02 = ? (Note: How many zeros are there after the decimal point?)	🌳	

Group C - T5 areatrap (questions 9-12)

#	Question	Difficulty	Working Space
R9	The upper base of the trapezoid is 6 cm, the lower base is 10 cm, and the height is 4 cm. Area = ?	🌱	
R10	Triangular base 8 cm, height 5 cm. Area = ? (Careful: Did you divide by 2?)	🌱	
R11	The parallelogram has a base of 9 cm, a height of 6 cm, and a hypotenuse of 7 cm. Area = ?	🌿	
R12	A combined figure consists of a trapezoid and a parallelogram. Trapezoid: (4+8)×3÷2, parallelogram: 5×3. Total area = ?	🌳	

Group D - T9 fraction operation (questions 13-16)

#	Question	Difficulty	Working Space
R13	$\frac{1}{3} + \frac{1}{6}$ =? (Get points first!)	🌱	
R14	$\frac{3}{4} - \frac{1}{6}$ =? (Remember to keep it simple!)	🌿	
R15	$1\frac{1}{2} + 2\frac{2}{3}$ =? (Tip: Convert Improper Fractions)	🌳	
R16	$\frac{2}{3} + \frac{1}{4} + \frac{1}{6}$ =? (Three fractions are combined)	🌳	

Group E — T3 reverse towardquestion (questions 17-20)

#	Question	Difficulty	Working Space
R17	Add 15 to a number and then multiply it by 3. The result is 90. Find this number. (Use backwards reasoning)	🌿	

R18	After Xiao Ming used the saved |||SEP|||, he deposited another $30, and now he has $90. How many yuan was it originally?$\frac{1}{4}$After you use it, you deposit another $30 and now you have $90. How many yuan was it originally?	🌳	
R19	If you subtract 28 from a number, divide it by 4, and then add 10, the result is 25. Find the original number.	🌳	
R20	There is some water in the bucket. After pouring out the |||SEP|||, add another 3 liters, now you have 10 liters. How many liters did it originally have?$\frac{2}{5}$Then, add another 3 liters and now you have 10 liters. How many liters did it originally have?	🏔	

V. Additional reinforcement of practice (total 22 questions · full trap mix)

Basic layer (question 21-28 · Everyone must do)

#	Question	Difficulty	Working Space
21	Round 12,345,678 to the nearest million.	🌱	
22	Calculate 0.4 × 0.7 = ?	🌱	
23	Triangular base 10 cm, height 6 cm. Area = ?	🌱	
24	$\frac{1}{5} + \frac{3}{10} = ?$	🌱	
25	Round 5,678,900 to the nearest hundred thousand digits.	🌱	
26	Calculate 0.8 × 0.05 = ?	🌿	
27	The upper base of the trapezoid is 5 cm, the lower base is 7 cm, and the height is 6 cm. Area = ?	🌱	
28	$\frac{2}{3} + \frac{1}{9} = ?$	🌱	

Advanced layer (question 29-36 · 🧗🚀 Select do)

#	Question	Difficulty	Working Space
29	Take 67,543,210 to the nearest tens of millions and write the Chinese pronunciation.	🌿	

30	Calculate 0.35 × 0.6 = ? (Multiply the integers first, then count the decimal places)	🌿	
31	The parallelogram has a base of 12 cm and a height of 5 cm. Area = ? (Don't use a bevel!)	🌿	
32	$\frac{3}{5} + \frac{2}{3}$ = ? (LCM(5,3)= ?)	🌿	
33	Three times a number plus 12 equals 48. Find a certain number. (Backward calculation: 48−12=36, 36÷3=?)	🌿	
34	Calculate 0.45 ÷ 0.5 = ? (Hint: divide by 0.5 = multiply by 2)	🌿	
35	The upper base of the trapezoid is 3 cm, the lower base is 9 cm, and the height is 8 cm. Area = ?	🌿	
36	$2\frac{1}{3} + 1\frac{1}{4}$=? (convert improper fractions first)	🌳	

🌳 challenge layer (question 37-42 · 🚀 choose do)

#	Question	Difficulty	Working Space
37	If you multiply a number by 0.5 and then add 8, the result is 23. Find the original number.	🌳	
38	A rectangle is 15 cm long and 8 cm wide. Cut out a triangle from it (base 8 cm, height 5 cm). Remaining area = ?	🌳	
39	$\frac{5}{8} + \frac{3}{4} + \frac{1}{2}$=? (Three numbers can be divided into three numbers, LCM(8,4,2)=?)	🌳	
40	For a barrel of oil, $\frac{1}{4}$ was used on the first day, and the remaining $\frac{1}{3}$ was used on the second day, leaving 12 liters. How many liters did it originally have? (Double reverse!)	⛰️	
41	Round 99,876,543 to the nearest million. And explain: why not 100,000,000?	🌳	
42	Calculate 0.125 × 0.8 = ? Write out the complete process of "first multiply integers → count decimal places → point decimal points".	🌳	

VI. The Lessoncorecommon errorsummary

🎯 **Review of Learning Objectives - After completing this lesson you should be able to:**

☐ Identify all trap types in our hall

☐ Solve 🌱basic questions independently (100% correct)

☐ Challenge 🌿 Advanced questions (80%+ correct) ☐ Explain the lesson formula to classmates

✅ **本堂自我檢查（完成後打剔）**

☐ 我識得分辨每個知識點嘅陷阱 ☐ 我能夠獨立完成🌱基礎題 ☐ 我能夠挑戰🌿進階題

☐ 我記得住口訣

#	common error	Correct Approach
1	**Take approximations to see misalignment**	Circle the target position first → look at the next digit → ≥5 and <5 will be rounded → clear the rest
2	**Wrong decimal point when multiplying decimals**	After multiplying the integers → count the total number of decimal places → click from right to left
3	**The area of the trapezoid is forgotten ÷2**	Remember the formula: (upper base + lower base) × height ÷ 2, not multiplied by 2
4	**Parallelogram uses hypotenuse**	Area = base × height (height is the vertical distance, not the hypotenuse)
5	**Direct addition of fractions with different denominators**	LCM must be found first through the denominator → the numerator is expanded simultaneously → the denominator remains unchanged
6	**The answer is not reduced**	Final step: Check if the numerator and denominator have common factors
7	**Reverse question inference**	Work backwards from the final result: addition, subtraction, multiplication, division, order reversal

🏠 家長角落 · Parent Corner

今日學咗咩？ 小朋友學咗「**Simulation 2 Review + Winter Vacation SSPA Special Training Plan**」。

最易錯嘅 3 個陷阱： 留意老師圈出嘅陷阱題目

你可以問小朋友： 「你可唔可以解釋「Simulation 2 Review + Winter Vacation SSPA Special Training Plan」嘅最重要口訣俾我聽？」

溫馨提示： 唔需要識教數學 — 只需要問小朋友「點解咁計？」同「有冇檢查陷阱？」就夠。

📝 **教師參考**：熱身題答案 → 1)___ 2)___ 3)___ 4)___ 5)___ | 課堂練習重點關注題號：🌳挑戰層 17-21